ASSIGNMENTS

# Problem set 3 - written part

AI504 — Knowledge Representation

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# Contents

# 1. Exercise 1.8

Here is a set $\Gamma$:

$$\Gamma = \{\text{All } a \text{ are } b, \text{All } c \text{ are } d, \text{All } a \text{ are } c, \text{All } a \text{ are } e, \text{All } c \text{ are } e\}$$

Then $\Gamma \not\models \text{All } d \text{ are } b$. The point of this problem is to give two models of $\Gamma$ where All d are b is false.

1. Find the canonical model of $\Gamma$, and check that All $d$ are $b$ is false in that model.
2. Find a model $\mathcal{M}$ with just one element such that $\mathcal{M} \models \Gamma$ but $\mathcal{M} \not\models \text{All } d \text{ are } b$ [Hint: You can do this by modifying the model in Exercise 1.8. That is, you use a model $\mathcal{M}$ with $M = \{*\}$, and with the interpretation function given by something like $(1.8)$. The only difference is that we don't want $y$ on the right, we want .... You can also get a one-point model this by using Exercise 1.9 just below. On the other hand, some people might find Exercise 1.9 easier to think about after working on this exercise.]

## 1.1. Solution

1. Let $M = \{a, b, c, d, e\}$ and $\Gamma = \{\text{All } a \text{ are } b, \text{All } c \text{ are } d, \text{All } a \text{ are } c, \text{All } a \text{ are } e, \text{All } c \text{ are } e\}$

   Then $\mathcal{M} = (M, [\![]\!] : M \to \mathcal{P}(M))$ becomes the canonical model, because the universe is the same as the signature, and it only satisfies the sentences in $\Gamma$.

$$\begin{aligned}
[\![a]\!] &= \{a\} \\
[\![b]\!] &= \{a, b\} \\
[\![c]\!] &= \{a, c\} \\
[\![d]\!] &= \{a, c, d\} \\
[\![e]\!] &= \{a, c, e\}
\end{aligned}$$

The different sentences that can be stated from this canonical model are:

$$\begin{array}{lll}
\mathcal{M} \models \text{All } a \text{ are } b & \mathcal{M} \models \text{All } c \text{ are } d & \mathcal{M} \models \text{All } a \text{ are } c \\
\mathcal{M} \models \text{All } a \text{ are } e & \mathcal{M} \models \text{All } c \text{ are } e & \mathcal{M} \not\models \text{All } d \text{ are } b
\end{array}$$

This model thus satisfies all sentences in $\Gamma$, and doesn't satisfy All $d$ are $b$.

2. Let's assume that M is still the signature $\{a, b, c, d, e\}$ and $\Gamma$ still holds. Then we can create a $\mathcal{M}$ that only contains one element $M = \{1\}$ and that $[\![d]\!] = \{1\}$ and every other nouns interpretation is simply the empty set.

$$[\![a]\!] = \emptyset$$
$$[\![b]\!] = \emptyset$$
$$[\![c]\!] = \emptyset$$
$$[\![d]\!] = \{1\}$$
$$[\![e]\!] = \emptyset$$

From this $\mathcal{M}$ we can see that $\mathcal{M} \models \Gamma$:

$$\mathcal{M} \models \text{All } a \text{ are } b$$
$$\mathcal{M} \models \text{All } c \text{ are } d$$
$$\mathcal{M} \models \text{All } a \text{ are } c$$
$$\mathcal{M} \models \text{All } a \text{ are } e$$
$$\mathcal{M} \models \text{All } c \text{ are } e$$

But since the empty set is a subset of every set, and not reverse, the model $\mathcal{M}$ doesn't satisfy the property:

$$\mathcal{M} \not\models \text{All } d \text{ are } b$$